CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 5: Quadratic Equations

Every formula and result of the chapter on one sheet

1	Standard form	$ax^2 + bx + c = 0,\ a \neq 0$
2	Zero product rule	If $pq = 0$, then $p = 0$ or $q = 0$
3	Quadratic formula	$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
4	Discriminant	$D = b^2 - 4ac$
5	Real and unequal roots	$D > 0$
6	Real and equal roots	$D = 0$; each root $= -\dfrac{b}{2a}$
7	Roots not real	$D < 0$
8	Real roots	$D \geq 0$
9	Sum of the roots	$-\dfrac{b}{a}$
10	Product of the roots	$\dfrac{c}{a}$
11	Equation with roots α and β	$x^2 - (\alpha + \beta)x + \alpha\beta = 0$